

Lecture 1, Mar 16 2026.

10.3 Polar Coordinates.

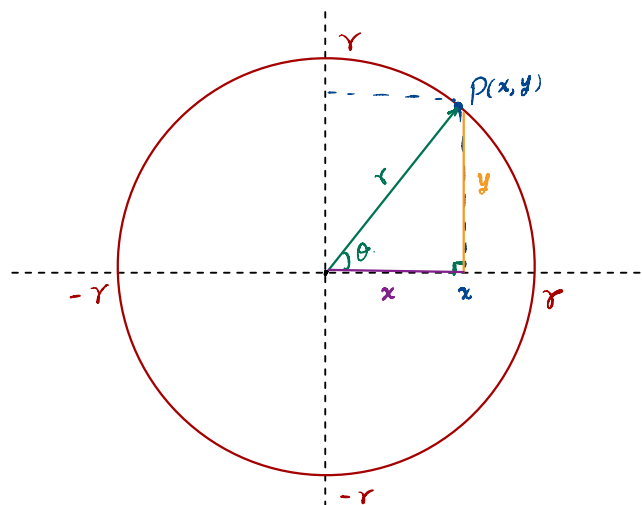
We know that if f, g are continuous functions then

$$C: \begin{cases} x = f(t) \\ y = g(t) \end{cases} \quad t \in [a, b].$$

defines a parametric curve. Let consider the equation:

$$x^2 + y^2 = r^2.$$

What is the parametric equations?



$$\sin \theta = \frac{y}{r}$$

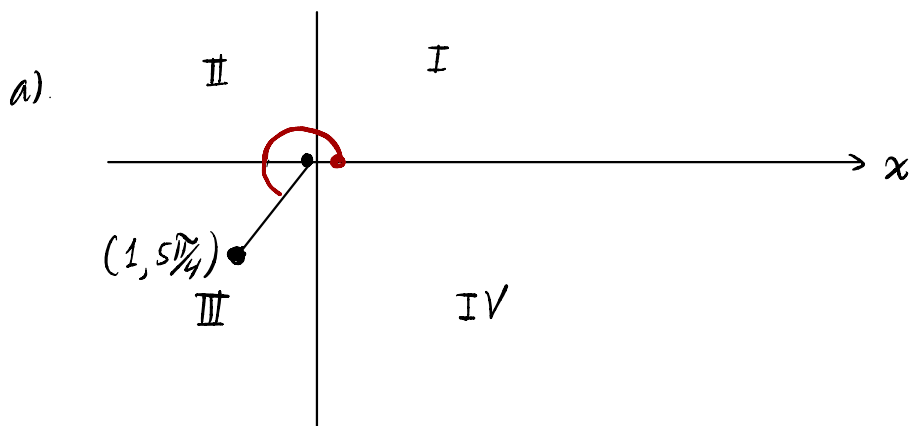
$$\cos \theta = \frac{x}{r} \quad \text{then}$$

$$x = r \cos \theta, \quad y = r \sin \theta.$$

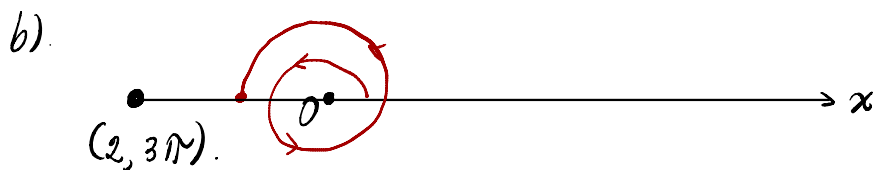
Example. Plot the points whose polar coordinates are given.

- a). $(1, 5\pi/4)$ b). $(2, 3\pi)$ c). $(2, -2\pi/3)$ d). $(-3, 3\pi/4)$.

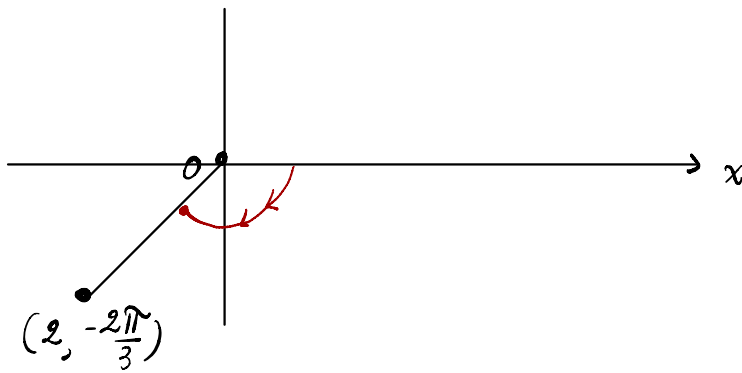
Sln.



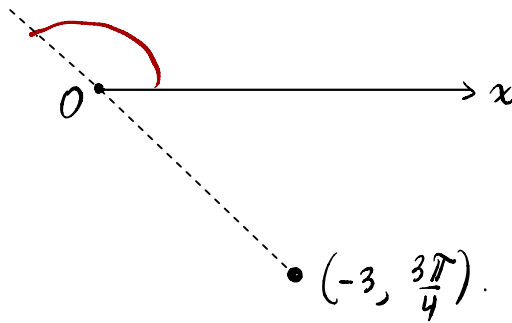
$$\frac{5\pi}{4} \longrightarrow 225^\circ$$



c).



d).



Then, from the parametric equations we have that

$$x^2 + y^2 = r^2, \quad x = r \cos \theta, \quad y = r \sin \theta; \quad \tan \theta = \frac{y}{x}$$

Example. Convert the point $(2, \pi/3)$ from polar to cartesian coordinates.

Sln. $r = 2$ $\theta = \frac{\pi}{3}$ Then

$$x = 2 \cos \frac{\pi}{3} = 2 \left(\frac{1}{2} \right) = 1.$$

$$y = 2 \sin \frac{\pi}{3} = 2 \left(\frac{\sqrt{3}}{2} \right) = \sqrt{3}.$$

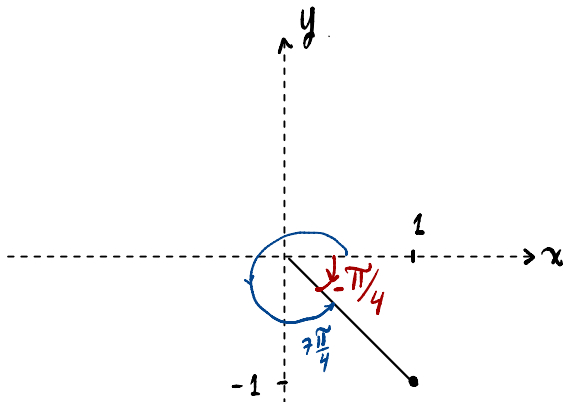
then, the point is $(1, \sqrt{3})$

Example. Represent the point $(1, -1)$ in terms of polar coordinates.

Sln. $r = \sqrt{x^2 + y^2} = \sqrt{1^2 + (-1)^2} = \sqrt{2}.$

$$\tan \theta = \frac{y}{x} = \frac{-1}{1} = -1 \quad \text{then} \quad \theta = \arctan(-1) = -\frac{\pi}{4}.$$

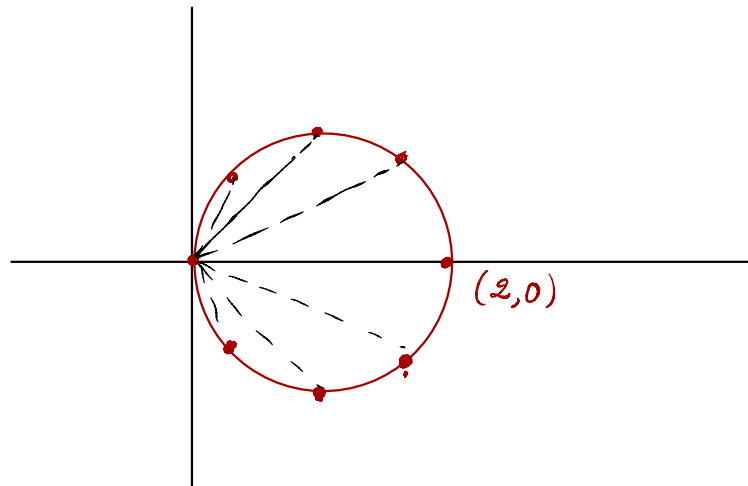
$$(1, -1) = (\sqrt{2}, -\frac{\pi}{4}) = (\sqrt{2}, \frac{7\pi}{4})$$



Example. a). Sketch the curve with polar equation $r = 2 \cos \theta$.
b). Find a cartesian equation for this curve.

Sln. a).

θ	$r = 2 \cos \theta$
0	2
$\pi/6$	$\sqrt{3}$
$\pi/4$	$\sqrt{2}$
$\pi/2$	0
$2\pi/3$	-1
$3\pi/4$	$-\sqrt{2}$



b). We know $x = r \cos \theta$. then $\cos \theta = \frac{x}{r}$ but $r = 2 \cos \theta$.

then $r = \frac{2x}{r}$ So, $r^2 = 2x$ and since

$$x^2 + y^2 = r^2 = 2x$$

$$x^2 + y^2 - 2x = 0.$$

Problem: Sketch the curve $r = \cos 2\theta$.

θ	$r = \cos 2\theta$	θ	$r = \cos 2\theta$
0	1	$\frac{3\pi}{4}$	0
$\frac{\pi}{4}$	0	π	1
$\frac{\pi}{2}$	-1	$-\frac{\pi}{4}$	0
		$-\frac{\pi}{2}$	-1

